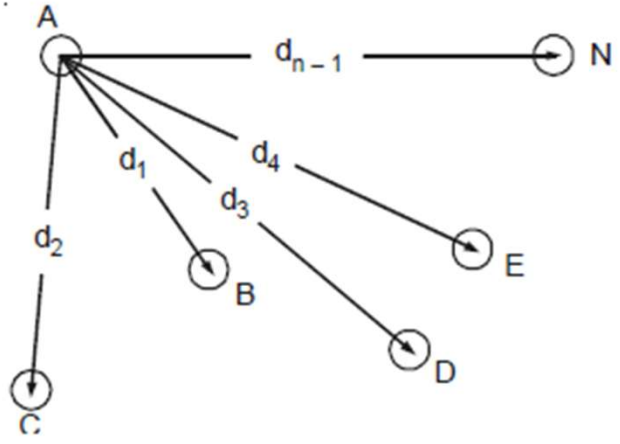
Potential at a Charged Conductor in a Group of Charged Conductors

Consider a group of long straight conductors A, B, C, D, E, ..., N having charges of q_1, q_2, q_3, q_4, q_5, ..., q_n coulombs per metre length respectively (Fig. 5.33).



$$V_{AN} = \frac{1}{2\pi \epsilon_0}\left[q_1 \log_e \frac{1}{r} + q_2 \log_e \frac{1}{d_1} + q_3 \log_e \frac{1}{d_2} + \ldots + q_n \log_e \frac{1}{d_{n-1}}\right] \text{ volts} \quad \ldots(5.39)$$